

tf.compat.v1.linalg.matmul Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/linalg/matmul

*Multiplies matrix a by matrix b , producing $a * b$.*

Function Signatures

```
tf.compat.v1.linalg.matmul( a, b, transpose_a=False,
transpose_b=False, adjoint_a=False, adjoint_b=False,
a_is_sparse=False, b_is_sparse=False, output_type=None,
grad_a=False, grad_b=False, name=None )
```

Code Examples

```
tf.compat.v1.linalg.matmul(  
a,  
b,  
transpose_a=False,  
transpose_b=False,  
adjoint_a=False,  
adjoint_b=False,  
a_is_sparse=False,  
b_is_sparse=False,  
output_type=None,  
grad_a=False,  
grad_b=False,  
name=None  
)
```

```
tf.compat.v1.linalg.matmul(  
a,  
b,  
transpose_a=False,  
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adjoint_a=False,  
adjoint_b=False,  
a_is_sparse=False,  
b_is_sparse=False,  
output_type=None,  
grad_a=False,  
grad_b=False,  
name=None  
)
```

```
a = tf.constant([1, 2, 3, 4, 5, 6], shape=[2, 3])  
a  # 2-D tensor  
  
b = tf.constant([7, 8, 9, 10, 11, 12], shape=[3, 2])  
b  # 2-D tensor  
  
c = tf.matmul(a, b)  
c  # `a` * `b`
```

```
a = tf.constant([1, 2, 3, 4, 5, 6], shape=[2, 3])
```

```
a # 2-D tensor
```

```
array([[1, 2, 3],
```

```
[4, 5, 6]], dtype=int32)>
```

Documentation

Multiplies matrix a by matrix b, producing $a * b$.

The inputs must, following any transpositions, be tensors of rank ≥ 2 where the inner 2 dimensions specify valid matrix multiplication dimensions, and any further outer dimensions specify matching batch size.

Both matrices must be of the same type. The supported types are: bfloat16, float16, float32, float64, int32, int64, complex64, complex128.

Either matrix can be transposed or adjointed (conjugated and transposed) on the fly by setting one of the corresponding flag to True. These are False by default.

If one or both of the matrices contain a lot of zeros, a more efficient multiplication algorithm can be used by setting the corresponding `a_is_sparse` or `b_is_sparse` flag to True. These are False by default. This optimization is only available for plain matrices (rank-2 tensors) with datatypes `bfloat16` or `float32`.

A simple 2-D tensor matrix multiplication:

A batch matrix multiplication with batch shape [2]:

Since python ≥ 3.5 the `@` operator is supported (see PEP 465). In TensorFlow, it simply calls the `tf.matmul()` function, so the following lines are equivalent:

Returns

`output[..., i, j] = sum_k (a[..., i, k] * b[..., k, j])`,
for all indices `i, j`.

`## Raises`